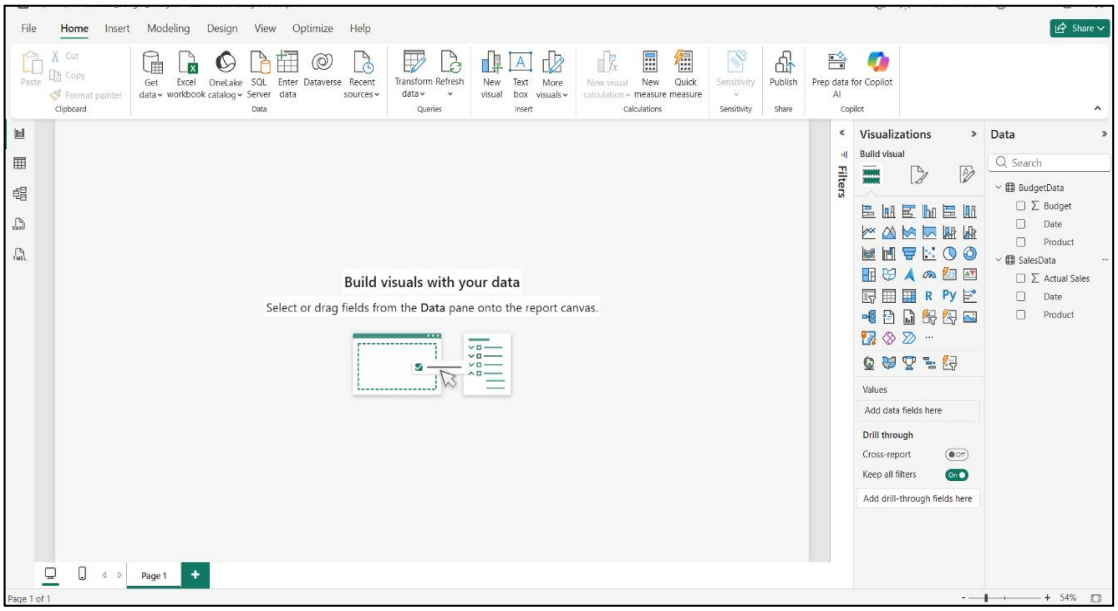
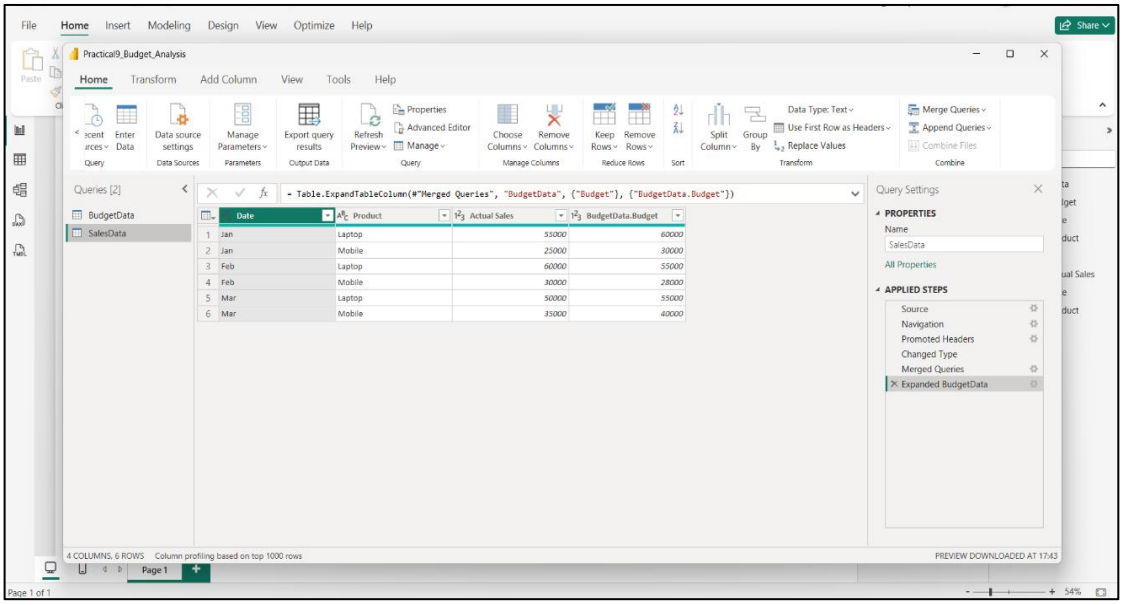
Practical9:	**Budget Analysis Dashboard** **Objective: Integrate sales and budget data to compare actual performance with targets**
Answer:	**Step 1: Import Data** Imported the sales and budget datasets into Power BI Desktop using Get Data → Excel Workbook.  **Step 2: Merge Datasets** Opened Power Query Editor and merged the Sales Data and Budget Data tables using the Date and Product columns with a Left Outer join. **Step 3: Expand Budget Column** Expanded the merged column to include the budget values in the sales table.  **Step 4: Apply Changes** Clicked Close & Apply to load the combined dataset into Power BI Desktop.

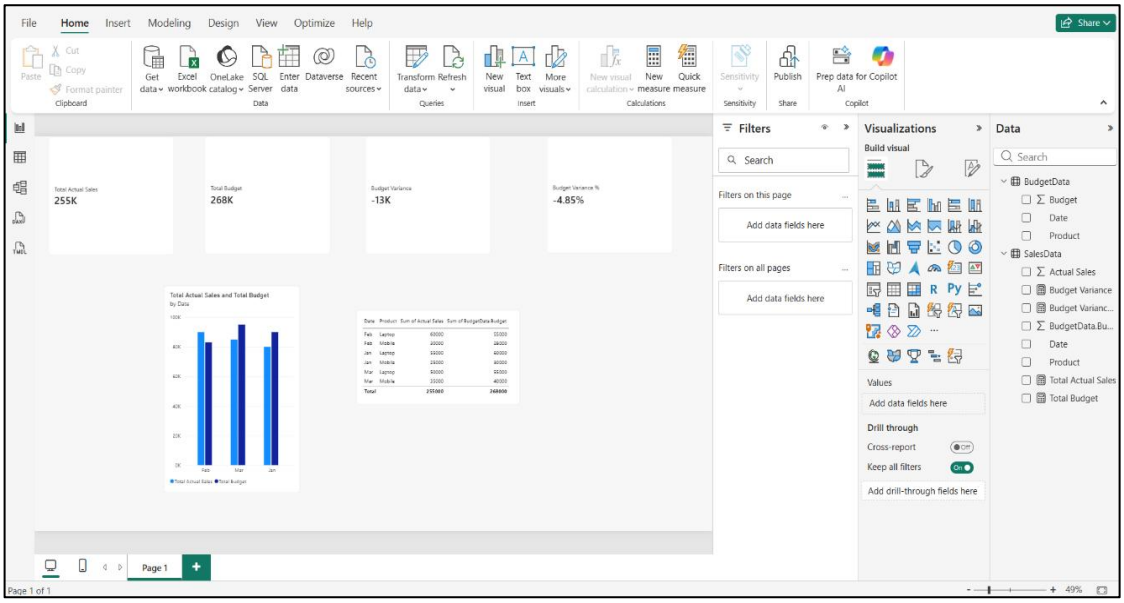
Step 5: Create Measures
Created measures to calculate Total Actual Sales, Total Budget, Budget Variance, and Budget Variance Percentage using DAX.

Step 6: Create KPI Cards
Created card visuals to display Total Actual Sales, Total Budget, Budget Variance, and Budget Variance Percentage.

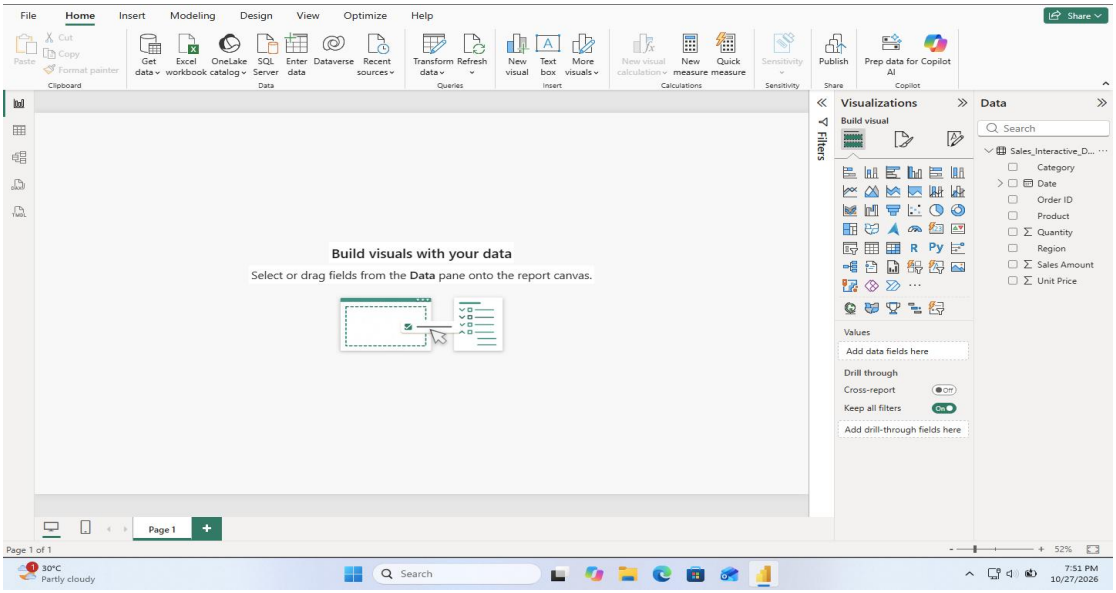
Step 7: Create Comparison Chart
Created a clustered column chart to compare actual sales with budget targets across different months.

Step 8: Create a Data Table
Created a table visual displaying Date, Product, Actual Sales, and Budget values.

Step 9: Arrange the Dashboard
Arranged the KPI cards, comparison chart, and data table to create a clear and informative budget analysis dashboard.



Result
Successfully integrated sales and budget data using Power Query and created a Budget Analysis Dashboard in Power BI. The dashboard displays key performance indicators and compares actual sales against budget targets.

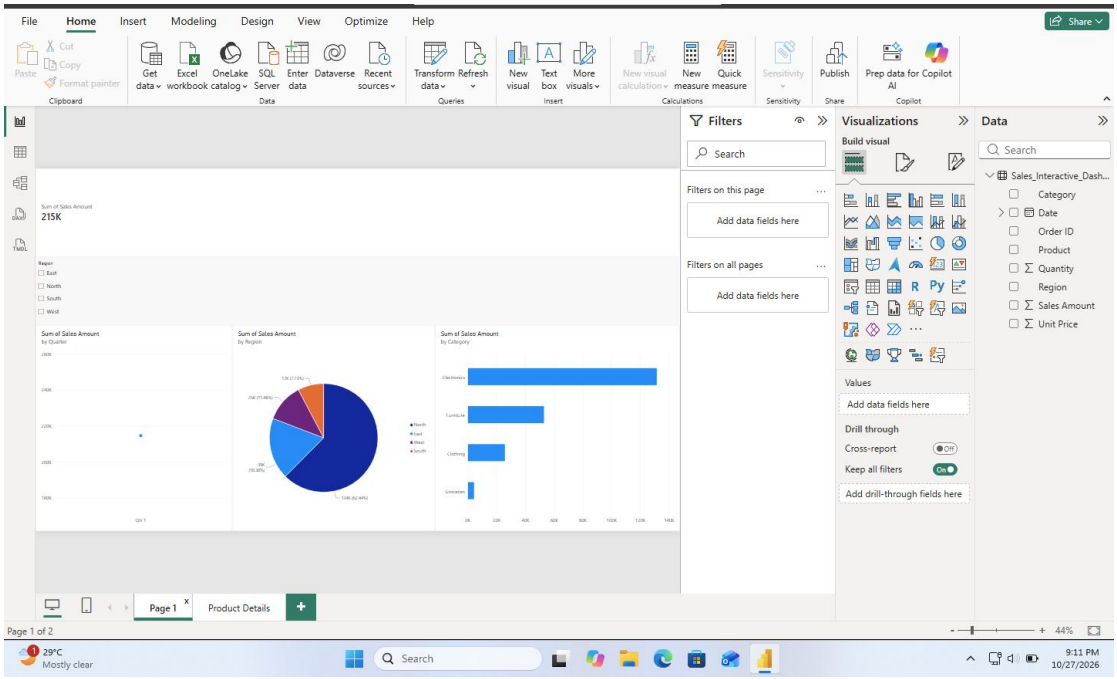
Practical10	**Interactive Dashboard Design** **Objective: Design an interactive dashboard using standard visualizations, slicers, and drill-through features.**
Answer:	**Step 1: Import the Dataset** 1. Open Power BI Desktop. 2. Click **Home → Get data → Text/CSV** 3. Select the downloaded CSV file. 4. Preview the dataset and click **Load.**  **Step 2: Create Standard Visualizations** **A. Bar Chart — Sales by Category.** 1. Open Report view. 2. Select the Clustered bar chart visual. 3. Drag Category to the Y-axis. 4. Drag Sales Amount to the X-axis. **B. Line Chart — Sales Over Time.** 1. Select the Line chart visual. 2. Drag Date to the X-axis. 3. Drag Sales Amount to the Y-axis. **C. Pie Chart — Sales by Region.** 1. Select the Pie chart visual. 2. Drag Region to the Legend field. 3. Drag Sales Amount to the Values field.

D. Card — Total Sales

1. Select the Card visual.
2. Drag Sales Amount into the card's data field.
3. Ensure the aggregation is set to Sum.

Step 3: Add a Region Slicer

1. Click an empty area of the report canvas.
2. Select the **Slicer** visual.
3. Drag **Region** into the slicer's field well.
4. Select a region, such as North or South.
5. Observe how the bar chart, line chart, pie chart, and card respond.



Step 4: Configure Drill-Through

1. Click the + icon at the bottom to create a new report page.
2. Rename the page **Product Details.**
3. Select the new page and locate the Drill-through field well in the Visualizations pane.
4. Drag **Product** into the Drill-through field well.
5. Add a Table visual to the page.
6. Add **Product, Category,** and **Sales Amount** to the table.
7. Return to the main dashboard.
8. Right-click a data point representing a product in a suitable visual.
9. Select **Drill through → Product Details**.

